GitHub Repository Blueprint: Strategic Portfolio Presentation

To turn your scripts, data visualizations, and maps into a professional, hiring-manager-grade GitHub repository, you must frame your files as a comprehensive **Intellectual Asset**. Recruiters and technical leads should not see isolated code snippets; they must immediately recognize an enterprise-ready **Environmental Intelligence Pipeline** that solves real-world data bottlenecks.

Below is the complete, production-ready markdown file generated for your repository root directory. It contains structural maps, operational runbooks, technical justifications, and explicit data findings.

📂 Recommended Repository Structure

Before pushing to GitHub, organize your local project folder (`C:\Users\markc\Documents\Data Science`) to match this clean layout. This structural discipline is highly valued by technical teams:

```
├── .gitignore                          # Prevents tracking large raw CSV assets and local DB binaries
├── README.md                           # Comprehensive operational runbook and portfolio documentation
├── requirements.txt                    # Explicit python package dependencies
├── factory_historian.db                # SQLite database layer with optimized coordinate indexing
│
├── scripts/
│   ├── scan_regions.py                # Metadata scanner isolating local administrative area nomenclature
│   ├── isolate_local.py               # Slices out the 45,360 regional records from the master file
│   ├── investigation.py               # Core anomaly engine isolating severe chemical/organic spikes
│   └── document_generator.py           # ReportLab script compiling the automated PDF audit report
│
└── outputs/
    ├── midlands_anomaly_map.png        # 300 DPI Matplotlib scatter plot with hotspot annotations
    ├── Critical_Pollution_Audit.txt    # Target facility text audit report
    ├── Water_Pipeline_Documentation.pdf # Professional system document report
    └── midlands_projected_qgis.qgs    # Saved QGIS project file with stylized raster risk heatmaps
```

📝 Step-by-Step GitHub Upload Protocol

Execute these commands sequentially in your VS Code terminal to initialize, link, and publish your project safely:

1. **Create a `.gitignore` File:**
   In your root directory, create a text file named `.gitignore` and add these two lines. This prevents git from uploading massive raw files or breaking local database connections:
   text
   ```
   *.csv
   *.db
   __pycache__/
   ```
   Use code with caution.
2. **Initialize Git & Commit Assets:**
   bash
   ```
   git init
   git add README.md requirements.txt scripts/ outputs/
   git commit -m "Initial commit: Deploying Midlands environmental data pipeline and QGIS mapping asset"
   ```
   Use code with caution.
3. **Link to GitHub & Push:**
   Create a new, empty repository on GitHub named `midlands-water-intelligence`, copy its remote URL, and link it:
   bash
   ```
   git remote add origin https://github.com
   git branch -M main
   git push -u origin main
   ```
   Use code with caution.

📊 Expanding Your Multi-Sector Polymath Portfolio

Now that your primary spatial intelligence pipeline is fully documented and structured for production deployment, we can choose the next high-value infrastructure project to build out your target portfolio segments:

- Would you like to build **Project 2: The Automated Industrial Predictive Maintenance Framework**? This project creates a machine learning model using Python to predict factory machinery failures, matching your heavy engineering experience at *Torres Pumps* and *Nexperia*.
- Or should we build **Project 3: The FMCG Demand Forecasting Web App**? This project creates a live browser dashboard using **Streamlit** to showcase time-series analysis and inventory optimization, matching your management experience at *B&Q* and *Focus DIY*.